

Alert Log files

⇒ The alert log file is a chronological log of messages and errors written out by an Oracle database.

⇒ This file should constantly be monitored to detect unexpected messages and corruptions.

⇒ Oracle will automatically create a new alert log file whenever the old one is deleted.

⇒ Also referred to as ALERT.LOG.
This file should constantly be monitored to detect unexpected messages and corruptions.

⇒ Alert log files in Oracle are like a diary for the database, keeping track of everything happens within it.

Locations Each database has an Alert-SID.log file.

- Specified by BACKGROUND_DUMP_DEST parameter.

- Resides on server with the database.

- Default location:

Oracle_BASE/diag/rdbms/DB-Name/SID/alert

Contents:

- Routine status messages and error conditions.
- Start up and shutdown messages.
- List of initialization parameters with non-default values.
- ALTER DATABASE or ALTER SYSTEM commands by the DBA.
- Operations on tablespaces and data files.
- Database errors.
- Warnings like ORA - errors.
- The alert log file contains the following information aid in tuning the database:
 - Time to perform archiving
 - Deadlock & timeout errors
 - Incomplete checkpoints
 - Instance recovery start and complete times.
- Administrative operations such as:
 - Create, alter, drop, database and tablespace statements

e Startup, Shutdown, archive log and recovery statements.

Errors messages related to:

- Functions of shared server
- Dispatcher process
- Errors during the automatic refresh of materialized view

c Messages will be written with timestamp.

c In Generic SQL use this command to find the location of the alert log:

⇒ SELECT

name

from v\$parameters

where name = 'background_dump_dest';

Format:

⇒ The alert log is in ASCII text format, making it easily readable.

Significance

- o The alert log serves as the primary source to diagnose issues when something goes wrong with the database.

Maintenance

- o Alert logs require periodic deletion or archiving to manage disk space efficiently.
- o Can be deleted or renamed, as it gets recreated.
- o Often archived daily by DBAs using batch jobs.

Examples of Recorded events

- o Corrupted redo logs
- o Tablespace space exhaustion

Background Trace Files

Trace file:

⇒ Trace file is a file that Oracle database creates to help diagnose and resolve operating problems.

⇒ Each server and background process writes to trace files.

⇒ When a process detects an internal error, it writes information about the error to its trace file.

Definition:

⇒ A trace file is generated during Oracle startup or when unexpected events occur in user or background processes.

⇒ It contains information like dumps of system global area, process global area, supervisor stack and registers.

Naming Conventions

- Trace file names include the instance name, process name and Oracle process numbers.

⇒ The file extension is typically TRC, as noted in the OS specific Oracle documentation.

Storage Location:-

⇒ BACKGROUND_DUMP_DEST stores trace files from Oracle background processes.

⇒ USER_DUMP_DEST for user trace files: (e.g. SQLDBA, SQLPLUS, PROXC).

Background Process Traces

- PMON; SMON traces an errors or instance recovery.
- QMON traces on spawning new processes.

User Session Traces:

- Created on errors (e.g. privilege issues) or by explicit command ALTER SESSION SET SQL_TRACE = TRUE.

- Useful for tuning SQL statements.

⇒ Trace files for user processes are created in two situations:

1. When some type of errors occur in a user session because of

a privilege problem, running out of space and so forth.

2. A trace file can be created explicitly with the command `alter session set sql_trace=true`.

Alert file Reference:

⇒ The Alert file indicates the location of trace files generated during internal error occurrences.

Formatting and Usage:

- Trace files may require formatting before use in diagnosing problems.
- The DUMPENT utility, available on most systems, is used for formatting trace files.
- It is specified in the OS-specific documentation.

Background Trace files

Definition:

- Background trace files are logs generated by background processes

in Oracle databases to provide detailed information about their activities and any encountered errors.

Purpose:

⇒ These trace files are essential for troubleshooting and diagnosing issues within the Oracle database environment, offering insights into background process activities and potential sources of errors.

Generation:

⇒ Each background process typically generates its own trace file containing information about its actions, executed SQL statements, errors encountered, and other relevant details.

Location:

⇒ Background trace files are stored in the directory specified by the BACKGROUND_DUMP_DEST initialization parameter.

Accessibility:

⇒ DBAs can access and examine these trace files using text editors or Oracle diagnostic tools.

Like ADRCI (Automatic Diagnostic Repository Command Interpreter).

Uses Trace files:

Definition

⇒ Uses trace files are logs generated when unexpected events occur within uses processes, such as SQLDBA, SQLPlus or Pro*C.

⇒ These files provide detailed information about the activities and errors encountered during the execution of uses processes.

Generation

⇒ Whenever an unexpected event occurs in a uses process, Oracle creates a trace file to capture relevant details.

⇒ These events could include errors, exceptions or other abnormal occurrences.

Content:

⇒ User trace files typically contain information such as:

- Details of errors or exceptions encountered.

- SQL statements executed prior to the error.

- State of relevant memory areas like the SGA and PGA.

- Stack traces to pinpoint the source of the error.

Location:

- User trace files are stored in the directory specified by the USER_DUMP_DEST initialization parameter.

Usage:

⇒ DBAs and developers can use trace files for:

- - Diagnosing and troubleshooting errors encountered during user operations.

- Identifying performance bottlenecks or inefficiencies in SQL queries.

o Understanding the sequence
of events leading up to an error
or exception.
